

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

September 2017

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper

Section B

Answer any **one** question on the separate Answer Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
Total	

This document consists of **13** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1** About one third of the injuries to racehorses involve tendon damage. In 2006, bone marrow stem cells were taken from injured racehorses and cultured so that they divided many times by mitosis. Each horse's cells were then injected into its damaged tendons. 80% of the treated horses returned to racing, compared with 30% of those treated conventionally.

- (a)** Adult stem cells such as these are described as multipotent.

Explain what is meant by the term *multipotent*.

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[1]

- (b)** In these treatment procedures, the bone marrow stem cells are stimulated to increase their rates of cell division so as to produce large enough numbers of cells to be injected into the horses.

Using your knowledge of cell signalling, describe how the rate of mitosis can be controlled.

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[4]

- (c)** Suggest how it is possible that bone marrow stem cells could differentiate into the range of cell types needed for repairing injuries.

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[2]

Like most mammals, T cell (T lymphocytes) in horses differentiate inside the thymus gland. During T cell differentiation, specific cell surface proteins known as CD proteins are produced and inserted into the cell surface membrane.

Fig. 1 shows the stages involved in the synthesis of a CD protein in a T cell.

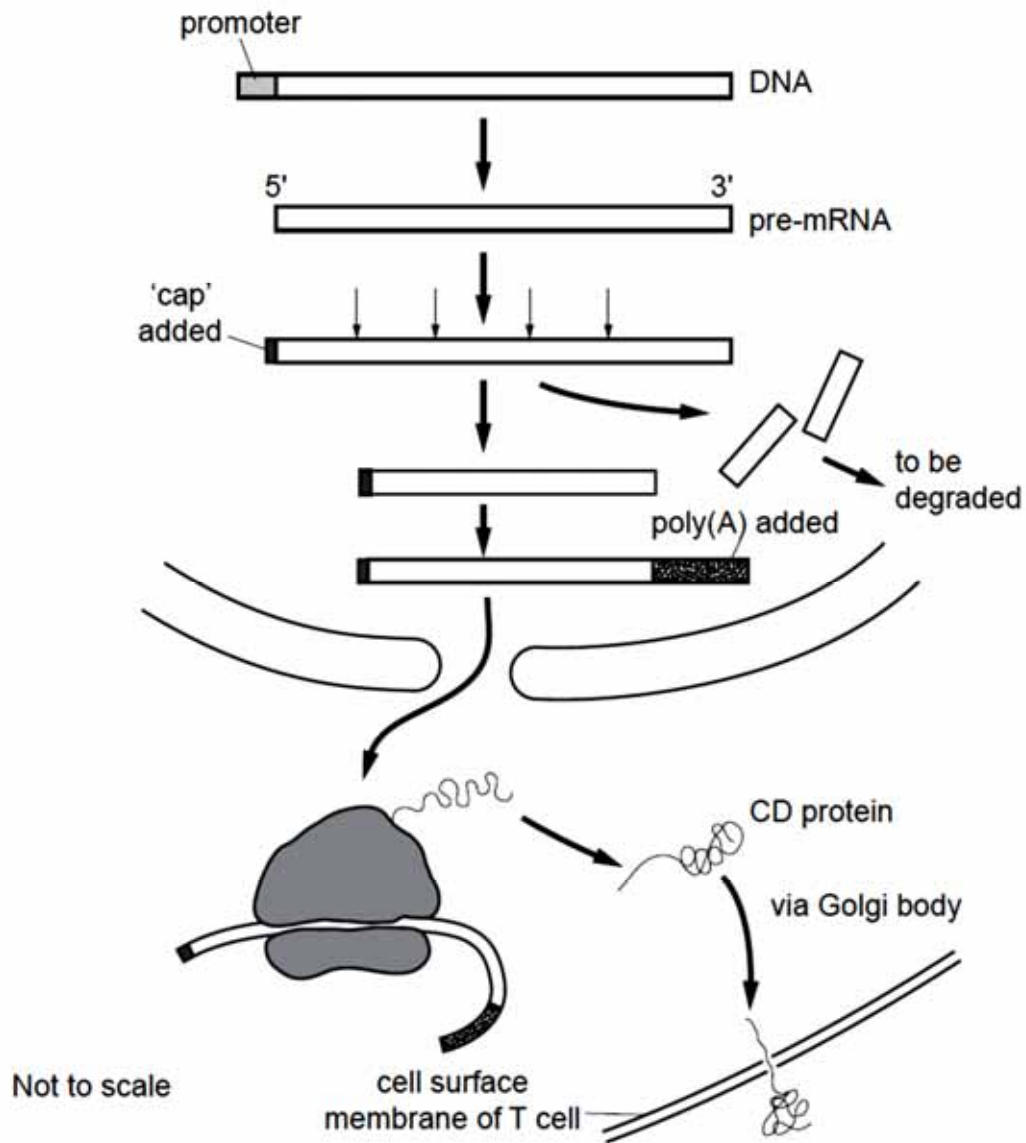


Fig. 1

(d) Explain the function of the promoter region of the DNA.

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[2]

- (e) Using **Fig. 1** as a guide, describe the events that occur in the nucleus of the T cell to produce a pre-mRNA encoding a CD protein.

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[3]

- (f) Explain why the cuts made in pre-mRNA are necessary for the T cell to produce a functional CD protein.

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[3]

- (g) Explain the functions for the 'cap' and the poly-A region attached to the mRNA.

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[2]

[Total: 17]

- 2 Lactate dehydrogenase (LDH) is an enzyme found in many organisms. Within the same organism, it can be found in different forms, called isoenzymes. The isoenzymes are structurally different but all catalyse the same reaction.

(a) **Fig. 2.1** shows a reaction catalysed by lactate dehydrogenase that occurs during anaerobic respiration in muscle tissue.

- (i) Complete **Fig. 2.1** by identifying the compounds **A**, **B** and **C**.

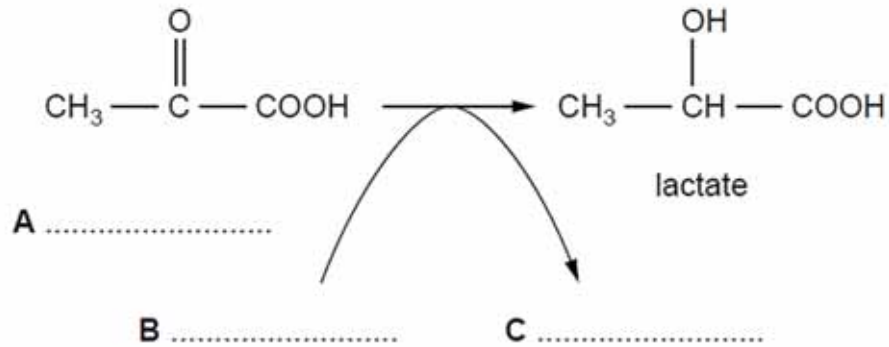


Fig. 2.1

[2]

- (ii) State where in the cell this reaction takes place.

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- (iii) Explain the importance of this reaction in mammalian muscle tissue.

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Lactate dehydrogenase isoenzymes are globular proteins, each consisting of four polypeptides.

- (b) Explain how the structure of an enzyme, such as lactate dehydrogenase is suited to its role.

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[4]

Lactate dehydrogenase isoenzymes are made up of two types of polypeptide:

- polypeptide M, which is coded for by the *LDH-A* gene and
- polypeptide H, which is coded for by the *LDH-B* gene.

Table 2 shows the composition of different human lactate dehydrogenase isoenzymes and examples of tissues and organs where each can be found.

Table 2

isoenzyme	polypeptide composition of enzyme	example of isoenzyme location
LDH-1	HHHH	heart red blood cells
LDH-2	HHHM	heart red blood cells
LDH-3	HHMM	brain lungs
LDH-4	HMMM	kidneys placenta
LDH-5	MMMM	liver skeletal muscles

- (c) With reference to **Table 2**, suggest how red blood cells of the same individual can produce different isoenzymes.

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[2]

Besides lactate dehydrogenase, another extensively studied protein which is also involved in respiration is cytochrome c. Cytochrome c plays an important role in oxidative phosphorylation as an electron carrier of the electron transport chain.

(d) **Fig. 2.2** shows the amino acid sequence of a section of the cytochrome c polypeptide chain retrieved from a human and the other species. The dashes shown in the figure indicates that the amino acid present at the position is identical to that of the human species.

Molecular homology of cytochrome c																							
		1				6				10				14			17	18		20			
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln	Cys	His	Thr	Val	Glu	Lys
Pig		-	-	-	-	-	-	-	-	-	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	-
Chicken		-	-	Ile	-	-	-	-	-	-	-	Val	Gln	-	-	-	-	-	-	-	-	-	-
Dogfish		-	-	-	-	-	-	-	-	Val	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	Asn
Drosophila	<<<	-	-	-	-	-	-	-	-	Leu	-	Val	Gln	Arg	-	Ala	-	-	-	-	-	-	Ala
Wheat	<<<	-	Asn	Pro	Asp	Ala	-	Ala	-	-	-	Lys	Thr	-	-	Ala	-	-	-	-	-	Asp	Ala
Yeast	<<<	-	Ser	Ala	Lys	-	-	Ala	Thr	Leu	-	Lys	Thr	Arg	-	Glu	Leu	-	-	-	-	-	-

Fig. 2.2

(i) Suggest what the data in **Fig. 2.2** indicate about the evolutionary relationships between humans and the other species.

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[3]

(ii) Explain why any such conclusions in (i) need to be treated with caution.

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[2]

- (e) Suggest how the differences in the amino acid sequences shown in **Fig. 2.2** may have come about.

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[2]

- (f) Suggest why the cytochrome c protein was chosen to compare amino acid sequences across the different species.

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[1]

[Total: 20]

- 3** Dengue fever (DF) is a disease caused by infection with a virus transmitted by the *Aedes aegypti* mosquito. It is an acute viral infection characterized by fever, rash, headache and muscle and joint pain. Occasionally, dengue virus infections progress to dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS).

There are 4 known serotypes of the disease-causing dengue virus. Individuals who become infected with one serotype obtain lifelong immunity against that serotype but not the other three.

Female *Aedes* mosquitoes are responsible for human-to-human transmission of the dengue virus. During blood feed, they acquire the proteins necessary for them to develop eggs. As a result of their short life cycle, they are able to multiply quickly, allowing dengue to spread.

- (a) (i)** State one reason why blood is a good source of protein.

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[1]

- (ii)** Outline the life cycle of the *Aedes aegypti*.

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[4]

Aedes aegypti originated in Africa but migrated to other continents via the slave trade in the 1500s and 1600s. The estimated range of the mosquito is primarily the tropics and sub-tropics but they have also been detected in countries with temperate climate such as parts of Europe and North America. As a result of the spread of *Aedes* vector across the globe, worldwide incidences of dengue have also been on the rise, posing huge public health concerns.

Fig. 3.1 shows the annual average number of both DF and DHF reported to the World Health Organisation (WHO) and the average annual number of countries reporting these cases from 1955 to 2007. The upward trend continues today.

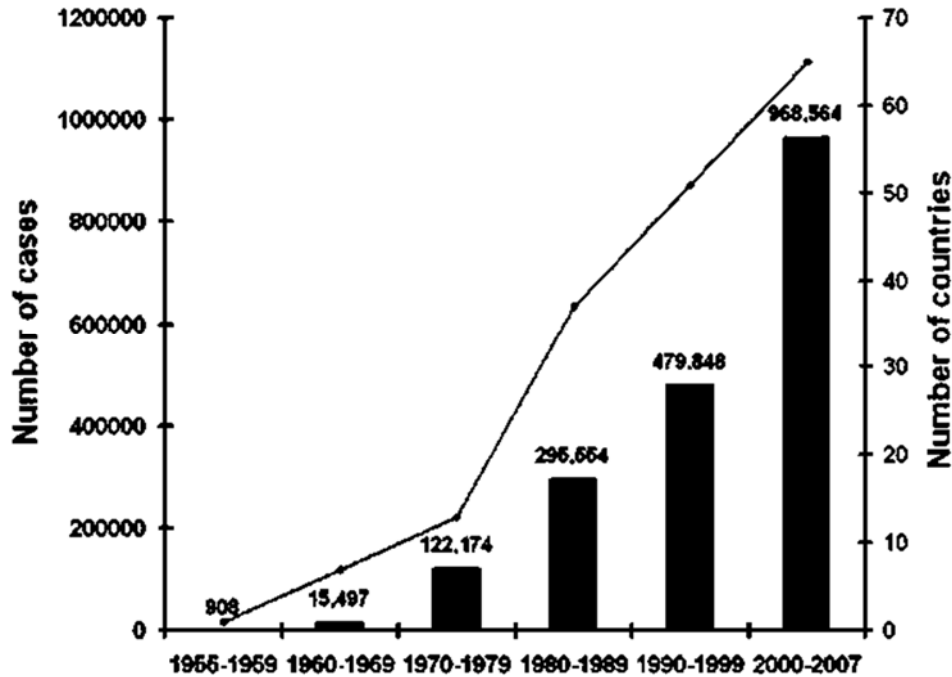


Fig. 3.1

While increased human population and global movement of people and cargo via air travel have undoubtedly assisted the spread of dengue, some scientists have attributed it to the effect of global warming.

Fig. 3.2 shows the global temperature departure ($^{\circ}\text{F}$) from the long-term average from 1880 to 2010. An additional trend line was plotted for the departure values between 1950 and 2010.

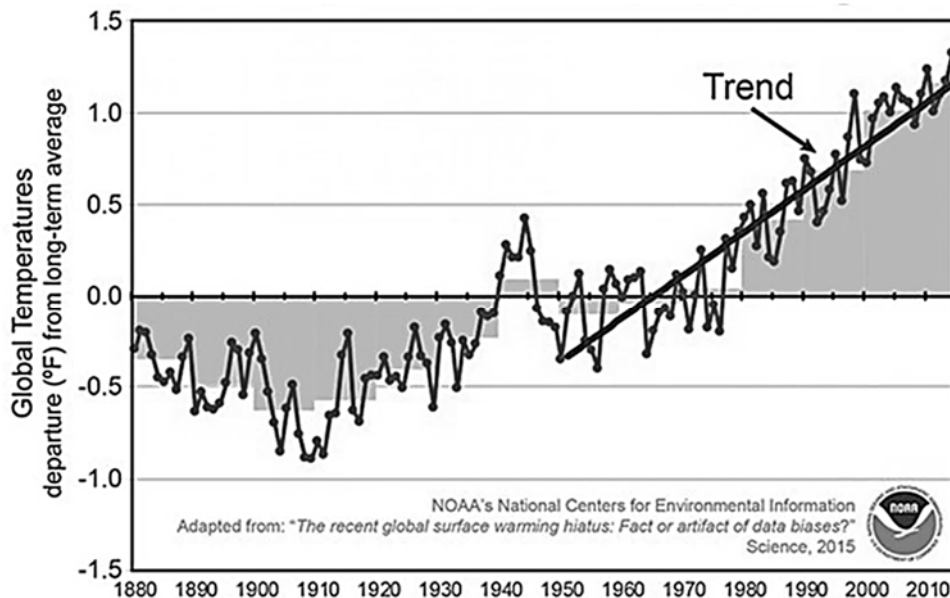


Fig. 3.2

- (b) Using evidence from **Fig 3.1** and **Fig. 3.2** and your own knowledge of global warming,
- (i) explain how the increase in global annual average number of DF / DHF could be caused by global warming.

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[3]

- (ii) suggest how global warming can lead to the spread of DF / DHF beyond the tropics into the temperate regions.

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[1]

- (c) *Wolbachia* is a natural bacterium present in up to 60% of insect species, but it is not usually found in the *Aedes aegypti* mosquito. Research has shown that when introduced into the *Aedes aegypti* mosquito, *Wolbachia* can stop the dengue virus from replicating inside the mosquito and hence prevent transmission to humans.

One possible way in which *Wolbachia* is used to suppress the *Aedes aegypti* mosquito population is through the release of male *Wolbachia-Aedes aegypti* mosquitoes. When these mosquitoes mate with the female wild-type *Aedes aegypti* that do not have *Wolbachia*, those females will lay eggs that do not hatch.

Country X released these male *Wolbachia-Aedes aegypti* mosquitoes in several housing estates as a form of vector control. Following three months of releases at the study site, the eggs were collected using ovitraps placed at several sites within the estate. The percentage of unhatched eggs were recorded for both the study and control sites.

- (i) State a statistical test that could have been used to determine whether the difference in the mean percentage of unhatched eggs between the study and control sites is significant.

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[1]

(ii) Suggest how a suitable control site was selected for the study.

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[1]

(iii) A summary of the results is shown in **Table 3**.

Table 3

mean percentage unhatched eggs		probability
study site	control site	
62.2	15.7	$p < 0.00001$

Comment on what these results show and suggest an explanation for any pattern.

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[2]

[Total: 13]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4(a) Outline the role of ATP in living organisms using named examples. [13]

(b) Discuss how the survival of species depends on DNA molecules being stable yet not *absolutely* stable. [12]

[Total: 25]

5(a) Outline how variation can arise in any population of living organisms. [13]

(b) Discuss the suggestion that all living organisms on Earth depend on phosphate. [12]

[Total: 25]